

COMPSCI 752

Big Data Management

Strategic Exercise 3 - Solutions

XQuery

Application domain. Consider the following XML document:

```
<?xml version="1.0" encoding="UTF-8"?>

<bib>
  <book year="1994">
    <title>TCP/IP Illustrated </title>
    <author>
      <last>Stevens</last>
      <first>W.</first>
    </author>
    <publisher>Addison-Wesley</publisher>
    <price>65.95</price>
  </book>
  <book year="1992">
    <title>Advanced Programming in the Unix environment</title>
    <author>
      <last>Stevens</last>
      <first>W.</first>
    </author>
    <publisher>Addison-Wesley</publisher>
    <price>65.95</price>
  </book>
  <book year="2000">
    <title>Data on the Web</title>
    <author>
      <last>Abiteboul</last>
      <first>Serge</first>
    </author>
    <author>
      <last>Buneman</last>
      <first>Peter</first>
    </author>
    <author>
      <last>Suciu</last>
      <first>Dan</first>
    </author>
    <publisher>Morgan Kaufmann Publishers</publisher>
    <price>39.95</price>
  </book>
  <book year="1999">
    <title>The Economics of Technology and Content for Digital TV</title>
    <editor>
      <last>Gerbarg</last>
      <first>Darcy</first>
      <affiliation>CITI</affiliation>
    </editor>
    <publisher>Kluwer Academic Publishers</publisher>
    <price>129.95</price>
  </book>
</bib>
```

Exercise 1 - XPath and Xquery.

Write the query “Return the names of all authors of books” in

- XPath
- XQuery
- and show the result when evaluated on the document above.

Solution.

- /bib/book/author
- let \$d := doc('bib.xml')
return <result>{\$d/bib/book/author}</result>

- <?xml version="1.0" encoding="UTF-8"?>
<result>
 <author>
 <last>Stevens</last>
 <first>W.</first>
 </author>
 <author>
 <last>Stevens</last>
 <first>W.</first>
 </author>
 <author>
 <last>Abiteboul</last>
 <first>Serge</first>
 </author>
 <author>
 <last>Buneman</last>
 <first>Peter</first>
 </author>
 <author>
 <last>Suciu</last>
 <first>Dan</first>
 </author>
</result>

xquery 中 变量一旦赋值
不能修改.

Exercise 2 - XPath and Xquery.

Write the query “Return the titles of all books published before 1997” in

- a. XPath
- b. XQuery
- c. and show the result when evaluated on the document above.

Solution.

- a. `/bib/book[@year < "1997"]/title`
- b.

```
for $bk in doc( bib.xml )/bib/book
  where $bk/@year < "1997"
  return $bk/title
```
- c.

```
<?xml version="1.0" encoding="UTF-8"?>
<title>TCP/IP Illustrated</title>
<title>Advanced Programming in the Unix environment</title>
```

Exercise 3 - Construction.

- a. Write in XQuery ‘Return year and title of all books published before 1997’
- b. and show the result when evaluated on the document above.

Solution.

- a.

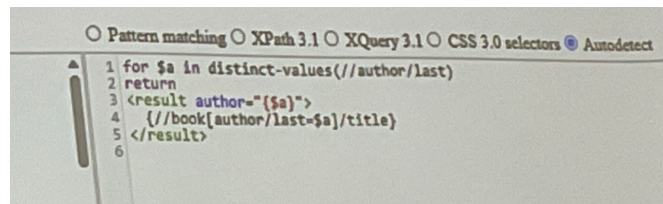
```
for $bk in doc( bib.xml )/bib/book
  where $bk/@year < "1997"
  return <book>{ $bk/@year, $bk/title }</book>
```
- b.

```
<?xml version="1.0" encoding="UTF-8"?>
<book year="1994">
  <title>TCP/IP Illustrated</title>
</book>
<book year="1992">
  <title>Advanced Programming in the Unix environment</title>
</book>
```

Exercise 4 - Grouping.

- a. Write in XQuery ‘Return titles for each author’
- b. and show the result when evaluated on the document above.

Solution.



- a.

```
<result>
{
  let $d:= doc("bib.xml")
  for $l in distinct-values($d//author/last)
  return
    <author name="{ $l }">
      { $d/bib/book[author/last= $l]/title }
    </author>
}
</result>
```
- b.

```
<?xml version="1.0" encoding="UTF-8"?>
<result>
  <author name="Stevens">
    <title>TCP/IP Illustrated</title>
    <title>Advanced Programming in the Unix environment</title>
  </author>
  <author name="Abiteboul">
    <title>Data on the Web</title>
  </author>
  <author name="Buneman">
    <title>Data on the Web</title>
  </author>
  <author name="Suciu">
    <title>Data on the Web</title>
  </author>
</result>
```

Exercise 5 - Aggregation.

- a. Write in XQuery ‘Return the average length of authors’ surnames for each book’
- b. and show the result when evaluated on the document above.

Solution.

- a.

```
for $b in doc('bib.xml')//book
let $a:=avg($b/author/string-length(last/text()))
return
  <book>
    {$b/title}
    <avg_surname>{$a}</avg_surname>
  </book>
```
- b.

```
<?xml version="1.0" encoding="UTF-8"?>
<book>
  <title>TCP/IP Illustrated</title>
  <avg_surname>7</avg_surname>
</book>
<book>
  <title>Advanced Programming in the Unix environment</title>
  <avg_surname>7</avg_surname>
</book>
<book>
  <title>Data on the Web</title>
  <avg_surname>7</avg_surname>
</book>
<book>
  <title>The Economics of Technology and Content for Digital TV</title>
  <avg_surname/>
</book>
```

Exercise 6 - Joins.

- a. Write in XQuery ‘Return pairs of books that share authors of the same name’.
- b. and show the result when evaluated on the document above.

Solution.

- a.

```
for $b1 in //book
  for $b2 in //book
    where $b1/title != $b2/title and $b1/author = $b2/author
      return
        <books>
          { $b1/title }
          { $b2/title }
        </books>
```
- b.

```
<?xml version="1.0" encoding="UTF-8"?>
<books>
  <title>TCP/IP Illustrated</title>
  <title>Advanced Programming in the Unix environment</title>
</books>
<books>
  <title>Advanced Programming in the Unix environment</title>
  <title>TCP/IP Illustrated</title>
</books>
```

Exercise 7 - Quantification.

- a. Write in XQuery ‘Return the books where no author has a first name with less than three characters’
- b. and show the result when evaluated on the document above.

Solution.

- a.

```
for $b in //book
where every $a in $b/author/first satisfies
  string-length($a)>2
return
  <book>
    {$b/title}
  </book>
```
- b.

```
<?xml version="1.0" encoding="UTF-8"?>
<book>
  <title>Data on the Web</title>
</book>
<book>
  <title>The Economics of Technology and Content for Digital TV</title>
</book>
```

Exercise 8 - Grouping and Aggregation.

- a. Write in XQuery ‘Return titles for each author, provided there are at least two.’
- b. and show the result when evaluated on the document above.

Solution.

- a.

```
<result>
  {
    let $d:= doc("bib.xml")
    for $l in distinct-values($d//author/last)
      where count(distinct-values($d//book[author/last= $l]/title))>=2
      return
        <author name="{ $l}">
          { $d/bib/book[author/last= $l]/title }
        </author>
  }
</result>
```
- b.

```
<?xml version="1.0" encoding="UTF-8"?>
<result>
  <author name="Stevens">
    <title>TCP/IP Illustrated</title>
    <title>Advanced Programming in the Unix environment</title>
  </author>
</result>
```

Exercise 9 - Interpreting and evaluating XQueries.

```
<bib>
  {
    for $b in doc("bib.xml")/bib/book
      where $b/publisher = "Addison-Wesley" and $b[@year > 1992]
      return
        <book year="{ $b/@year }">
          { $b/title }
        </book>
  }
</bib>
```

- a. What does the XQuery query above do?
- b. What is the result when evaluating the XQuery on the document above?

Solution.

- a. Return the title and year of publication for each book published by Addition-Wesley after 1992.
- b.

```
<?xml version="1.0" encoding="UTF-8"?>
<bib>
  <book year="1994">
    <title>TCP/IP Illustrated</title>
  </book>
</bib>
```

Exercise 10 - Interpreting and evaluating XQueries.

```
<results>
{
  for $b in doc("bib.xml")/bib/book,
    $t in $b/title,
    $a in $b/author/last
  return
    <result>
      { $t }
      { $a }
    </result>
}
</results>
```

- a. What does the XQuery query above do?
- b. What is the result when evaluating the XQuery on the document above?

Solution.

- a. Return each pair of a book title with one of its authors.

- b. `<?xml version="1.0" encoding="UTF-8"?>`

```
<results>
  <result>
    <title>TCP/IP Illustrated</title>
    <last>Stevens</last>
  </result>
  <result>
    <title>Advanced Programming in the Unix environment</title>
    <last>Stevens</last>
  </result>
  <result>
    <title>Data on the Web</title>
    <last>Abiteboul</last>
  </result>
  <result>
    <title>Data on the Web</title>
    <last>Buneman</last>
  </result>
  <result>
    <title>Data on the Web</title>
    <last>Suciu</last>
  </result>
</results>
```

Exercise 11 - Interpreting and evaluating XQueries.

```
<bib>
{
  for $b in doc("bib.xml")//book
    where $b/publisher = "Addison-Wesley" and $b/@year > "1991"
    order by $b/title
    return <book> { $b/@year } { $b/title } </book>
}
</bib>
```

- a. What does the XQuery query above do?
- b. What is the result when evaluating the XQuery on the document above?

Solution.

- a. For each book published by Addition-Wesley after 1991, return the year of publication and book title in alphabetic order of the title.
- b.

```
<?xml version="1.0" encoding="UTF-8"?>
<bib>
  <book year="1992">
    <title>Advanced Programming in the Unix environment</title>
  </book>
  <book year="1994">
    <title>TCP/IP Illustrated</title>
  </book>
</bib>
```